

34. Watchdog Timer (WDTA)

The watchdog timer (WDT) is a 14-bit down-counter. It can be used to reset this MCU when the counter underflows because its value cannot be refreshed due to the system being out of control.

In addition, a non-maskable interrupt can be generated by an underflow.

The refresh-permitted period can be set to refresh the counter and used as the condition to detect when the system runs out of control.

In this section, “PCLK” is used to refer to PCLKB.

34.1 Overview

Table 34.1 lists the specifications of the WDT and Figure 34.1 shows a block diagram of the WDT.

Table 34.1 WDT Specifications

Item	Specifications
Count source	Peripheral module clock (PCLK)
Clock division ratio	Divide by 4, 64, 128, 512, 2048, or 8192
Counter operation	Counting down using a 14-bit down-counter
Conditions for starting the counter	- Auto-start mode: Counting automatically starts after a reset is released - Register start mode: Counting is started by refresh operation (writing 00h and then FFh to the WDTRR register)
Conditions for stopping the counter	- Reset - In low power consumption states - A counter underflows or a refresh error occurs (only in register start mode)
Window function	Window start and end positions can be specified (refresh-permitted and refresh-prohibited periods)
Watchdog timer Reset sources	- Down-counter underflows - Refreshing outside the refresh-permitted period (refresh error)
Non-maskable interrupt/interrupt sources	- Down-counter underflows - Refreshing outside the refresh-permitted period (refresh error)
Reading the counter value	The down-counter value can be read by the WDTSR register.

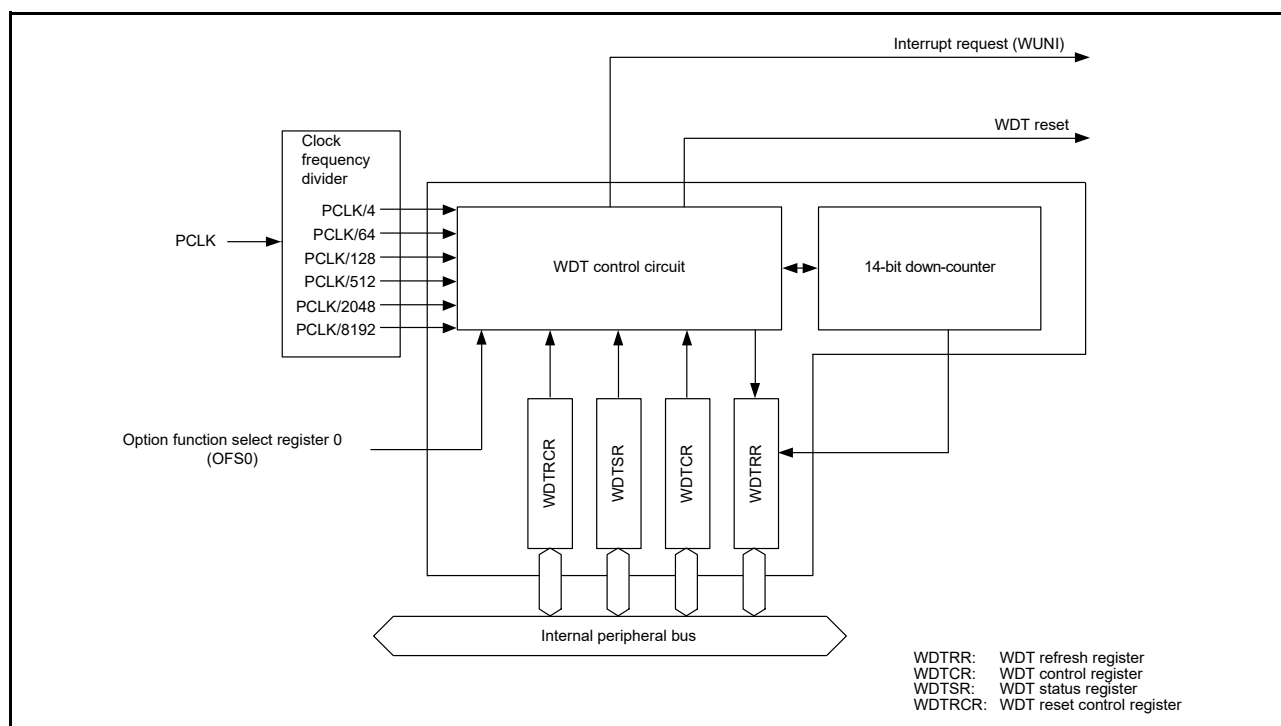


Figure 34.1 WDT Block Diagram

34.2 Register Descriptions

34.2.1 WDT Refresh Register (WDTRR)

Address(es): 0008 8020h



Bit	Description	R/W
b7 to b0	The down-counter is refreshed by writing 00h and then writing FFh to this register	R/W

WDTRR refreshes the down-counter of the WDT.

The down-counter of the WDT is refreshed by writing 00h and then writing FFh to WDTRR (refresh operation) within the refresh-permitted period.

After the down-counter has been refreshed, it starts counting down from the value selected by setting the WDTTOPS[1:0] bits in option function select register 0 (OFS0) in auto-start mode. In register start mode, counting down starts from the value selected by setting the WDTCR.TOPS[1:0] bits.

When 00h is written, the read value is 00h, when a value other than 00h is written, the read value is FFh.

For details of the refresh operation, refer to [section 34.3.3, Refresh Operation](#).

34.2.2 WDT Control Register (WDTCR)

Address(es): 0008 8022h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	RPSS[1:0]	—	—	RPES[1:0]	—	—	CKS[3:0]	—	—	—	—	—	TOPS[1:0]	—
0	0	1	1	0	0	1	1	1	1	1	1	0	0	1	1

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	TOPS[1:0]	Timeout Period Selection	b1 b0 0 0: 1024 cycles (03FFh) 0 1: 4096 cycles (0FFFh) 1 0: 8192 cycles (1FFFh) 1 1: 16384 cycles (3FFFh)	R/W
b3, b2	—	Reserved	These bits are read as 0 and cannot be modified.	R
b7 to b4	CKS[3:0]	Clock Division Ratio Selection	b7 b4 0 0 0 1: Divide-by-4 0 1 0 0: Divide-by-64 1 1 1 1: Divide-by-128 0 1 1 0: Divide-by-512 0 1 1 1: Divide-by-2048 1 0 0 0: Divide-by-8192 Setting other than above are prohibited.	R/W
b9, b8	RPES[1:0]	Window End Position Selection	b9 b8 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (window end position is not specified)	R/W
b11, b10	—	Reserved	These bits are read as 0 and cannot be modified.	R
b13, b12	RPSS[1:0]	Window Start Position Selection	b13 b12 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (window start position is not specified)	R/W
b15, b14	—	Reserved	These bits are read as 0 and cannot be modified.	R

There are some restrictions on writing to the WDTCR register. For details, refer to section 34.3.2, Control over Writing to the WDTCR and WDTRCR Registers.

In auto-start mode, the settings in the WDTCR register are disabled, and the settings in option function select register 0 (OFS0) are enabled. The bit setting made to the WDTCR register can also be made in OFS0 register. For details, refer to section 34.3.7, Correspondence between Option Function Select Register 0 (OFS0) and WDT Registers.

TOPS[1:0] Bits (Timeout Period Selection)

These bits select the timeout period (period until the down-counter underflows) from among 1024, 4096, 8192, and 16384 cycles, taking the divided clock specified by the CKS[3:0] bits as one cycle.

After the down-counter is refreshed, the combination of the CKS[3:0] and TOPS[1:0] bits determines the time (number of PCLK cycles) until the counter underflows.

Relations between the CKS[3:0] and TOPS[1:0] bit settings, the timeout period, and the number of PCLK cycles are listed in Table 34.2.

Table 34.2 Timeout Period Settings

CKS[3:0] Bits				TOPS[1:0] Bits		Clock Division Ratio	Timeout Period (Number of Cycles)	Cycles of PCLK Clock
b7	b6	b5	b4	b1	b0			
0	0	0	1	0	0	Divide-by-4	1024	4096
				0	1		4096	16384
				1	0		8192	32768
				1	1		16384	65536
0	1	0	0	0	0	Divide-by-64	1024	65536
				0	1		4096	262144
				1	0		8192	524288
				1	1		16384	1048576
1	1	1	1	0	0	Divide-by-128	1024	131072
				0	1		4096	524288
				1	0		8192	1048576
				1	1		16384	2097152
0	1	1	0	0	0	Divide-by-512	1024	524288
				0	1		4096	2097152
				1	0		8192	4194304
				1	1		16384	8388608
0	1	1	1	0	0	Divide-by-2048	1024	2097152
				0	1		4096	8388608
				1	0		8192	16777216
				1	1		16384	33554432
1	0	0	0	0	0	Divide-by-8192	1024	8388608
				0	1		4096	33554432
				1	0		8192	67108864
				1	1		16384	134217728

CKS[3:0] Bits (Clock Division Ratio Selection)

These bits specify the division ration of the clock used for the down-counter. The division ration can be selected from among the peripheral module clock (PCLK) divided by 4, 64, 128, 512, 2048, and 8192. Combined with the TOPS[1:0] bit setting, a count period between 4,096 and 134,217,728 cycles of the PCLK clock can be selected for the WDT.

RPES[1:0] Bits (Window End Position Selection)

These bits specify the window end position that indicates the refresh-permitted period. 75%, 50%, 25%, or 0% of the timeout period can be selected for the window end position. The selected window end position should be a value smaller than the window start position (window start position > window end position). If the window end position is greater than the window start position, only the window start position setting is enabled.

RPSS[1:0] Bits (Window Start Position Selection)

These bits specify the window start position that indicates the refresh-permitted period. 25%, 50%, 75%, or 100% of the timeout period can be selected for the window end position. The window start position should be set to a value greater the window end position. If the window start position is set to a value smaller than or equal to the window end position, the window end position is set to 0%.

Table 34.3 lists the counter values for the window start and end positions and Figure 34.2 shows the refresh-permitted period set by the RPSS[1:0], RPES[1:0], and TOPS[1:0] bits.

Table 34.3 Relationship between Timeout Period and Window Start and End Counter Values

TOPS[1:0] Bits		Timeout Period		Window Start and End Counter Value			
		Cycles	Counter Value	100%	75%	50%	25%
0	0	1024	03FFh	03FFh	02FFh	01FFh	00FFh
0	1	4096	0FFFh	0FFFh	0BFFh	07FFh	03FFh
1	0	8192	1FFFh	1FFFh	17FFh	0FFFh	07FFh
1	1	16384	3FFFh	3FFFh	2FFFh	1FFFh	0FFFh

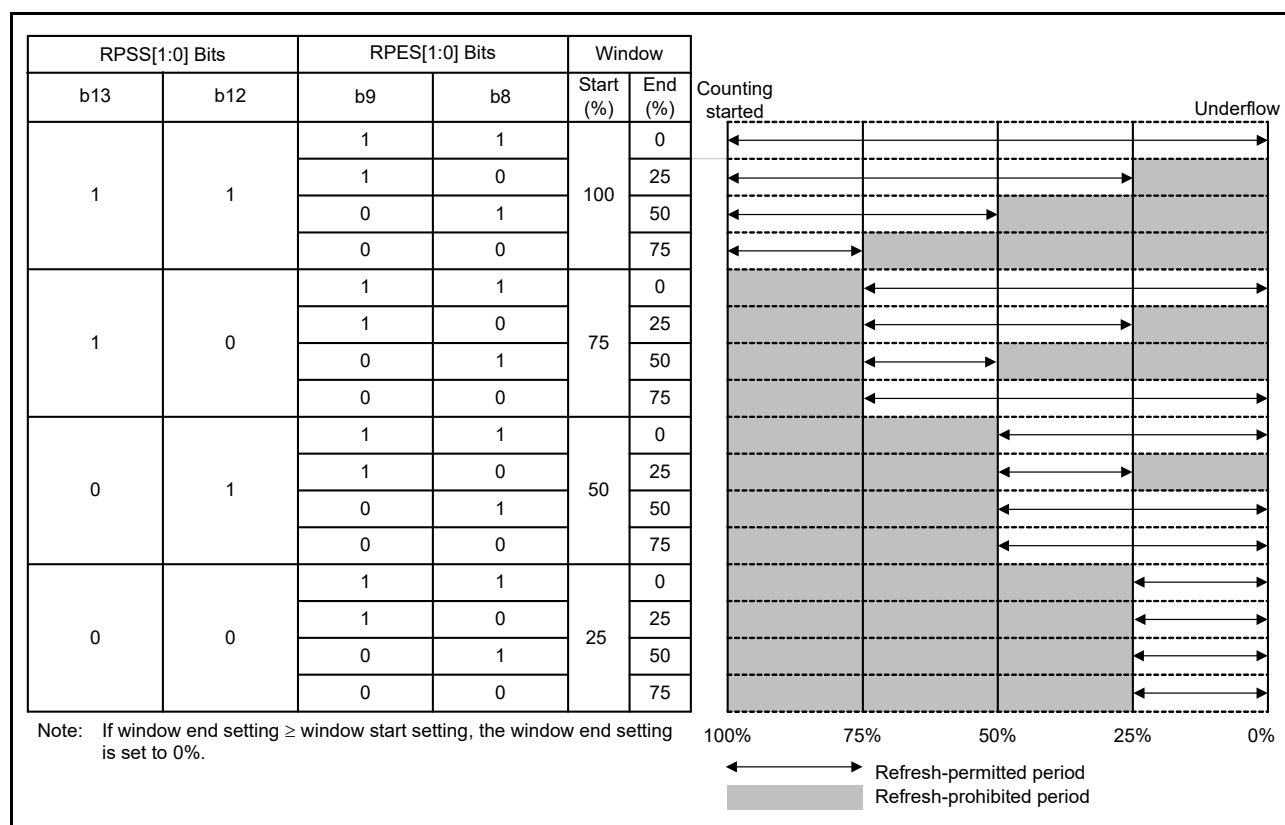


Figure 34.2 RPSS[1:0] and RPES[1:0] Bit Settings and the Refresh-Permitted Period

34.2.3 WDT Status Register (WDTSR)

Address(es): 0008 8024h



Bit	Symbol	Bit Name	Description	R/W
b13 to b0	CNTVAL[13:0]	Down-Counter Value	Value counted by the down-counter	R
b14	UNDF	Underflow Flag	0: No underflow occurred 1: Underflow occurred	R(/W) *1
b15	REFEF	Refresh Error Flag	0: No refresh error occurred 1: Refresh error occurred	R(/W) *1

Note 1. Only 0 can be written to clear the flag.

CNTVAL[13:0] Bits (Down-Counter Value)

Read these bits to confirm the value of the down-counter, but note that the read value may differ from the actual count by a value of one count.

UNDF Flag (Underflow Flag)

Read this flag to confirm whether or not an underflow has occurred in the down-counter.

The value 1 indicates that the down-counter has underflowed. The value 0 indicates that the down-counter has not underflowed.

Write 0 to the UNDF flag to set the value to 0. Writing 1 has no effect.

REFEF Flag (Refresh Error Flag)

Read this flag to confirm whether or not a refresh error (performing a refresh operation during a refresh-prohibited period) has occurred.

The value 1 indicates that a refresh error has occurred. The value 0 indicates that no refresh error has occurred.

Write 0 to the REFEF flag to set the value to 0. Writing 1 has no effect.

34.2.4 WDT Reset Control Register (WDTRCR)

Address(es): 0008 8026h

	b7	b6	b5	b4	b3	b2	b1	b0
	RSTIR QS	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 0 and cannot be modified.	R
b7	RSTIRQS	Reset Interrupt Request Selection	0: Non-maskable interrupt request or interrupt request output is enabled 1: Reset output is enabled	R/W

There are some restrictions on writing to the WDTRCR register. For details, refer to [section 34.3.2, Control over Writing to the WDTCR and WDTRCR Registers](#).

In auto-start mode, the WDTRCR register settings are disabled, and the settings in option function select register 0 (OFS0) are enabled. The bit setting made to the WDTCR register can also be made in the OFS0 register. For details, refer to [section 34.3.7, Correspondence between Option Function Select Register 0 \(OFS0\) and WDT Registers](#).

34.2.5 Option Function Select Register 0 (OFS0)

For details on the OFS0 register, refer to [section 34.3.7, Correspondence between Option Function Select Register 0 \(OFS0\) and WDT Registers](#).

34.3 Operation

34.3.1 Count Operation in Each Start Mode

The WDT has two start modes: auto-start mode, in which counting automatically starts after a reset is released, and register start mode, in which counting is started by refresh operation (writing to the register).

In auto-start mode, counting automatically starts after a reset is released in accordance with the settings in option function select register 0 (OFS0) in the ROM.

In register start mode, counting is started by refresh operation (writing to the register) after the respective registers are set after a reset is released.

Select auto-start mode or register start mode by setting the OFS0.WDTSTRT bit.

When the auto-start mode is selected, the settings in the WDTCCR and WDTRCR registers are disabled, and the settings in the OFS0 register are enabled.

On the other hand, when the register start mode is selected, the setting of the OFS0 register is disabled, and the settings of the WDTCCR and WDTRCR registers are enabled.

34.3.1.1 Register Start Mode

When the OFS0.WDTSTRT bit is 1, register start mode is selected, and the WDTCCR and WDTRCR registers are enabled.

After a reset is released, set the clock division ratio, window start and end positions, and timeout period in the WDTCCR register, and the reset output or interrupt request output in the WDTRCR register. Then, refresh the down-counter to start counting down from the value set by the WDTCCR.TOPS[1:0] bits.

Thereafter, as long as the counter is refreshed in the refresh-permitted period, the value in the counter is re-set each time the counter is refreshed and counting down continues. The WDT does not output the reset signal as long as this continues. However, if the down-counter underflows because the down-counter cannot be refreshed due to a program runaway, or if a refresh error occurs because the counter was refreshed outside the refresh-permitted period, the WDT outputs a reset signal or a non-maskable interrupt request/interrupt request (WUNI). Reset output or interrupt request output can be selected by setting the WDTRCR.RSTIRQS bit.

Figure 34.3 shows an example of operation under the following conditions.

- Register start mode (OFS0.WDTSTRT = 1)
- Reset output is enabled (WDTRCR.RSTIRQS = 1)
- The window start position is 75% (WDTCCR.RPSS[1:0] = 10b)
- The window end position is 25% (WDTCCR.RPES[1:0] = 10b)

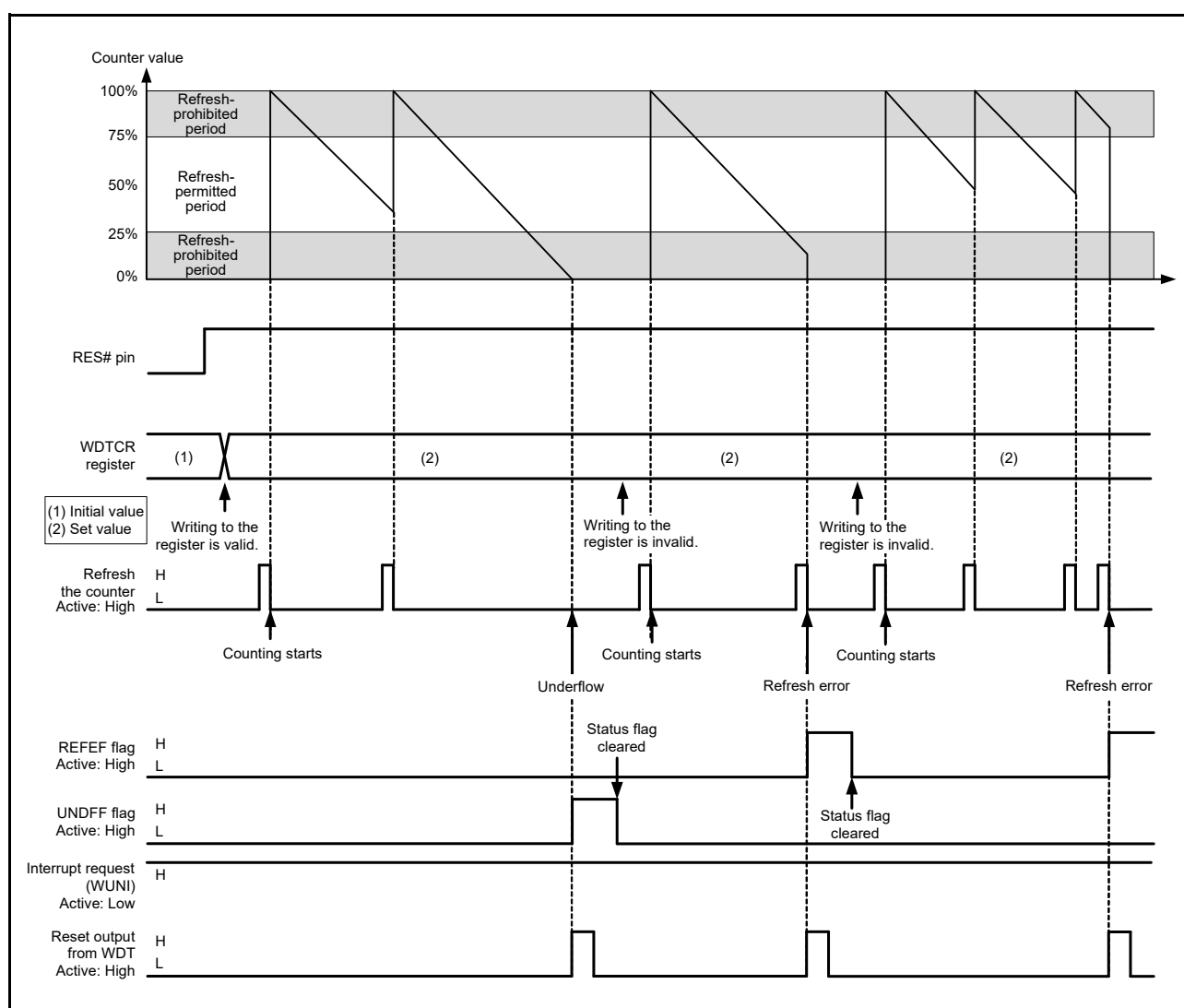


Figure 34.3 Operation Example in Register Start Mode

34.3.1.2 Auto-Start Mode

When the WDTSTRT bit in option function select register 0 (OFS0) is 0, auto-start mode is selected, the WDTCR and WDTRCR registers are disabled, and the settings in the OFS0 register are enabled.

Within the reset state, the setting values (clock division ratio, window start and end positions, timeout period, and reset output or interrupt request) of the OFS0 register are set in the WDT registers.

When the reset is released, the down-counter automatically starts counting down from the value set by the OFS0.WDTPS[1:0] bits.

After that, as long as the counter is refreshed in the refresh-permitted period, the value in the counter is re-set each time the counter is refreshed and counting down continues. The WDT does not output the reset signal as long as this continues.

However, if the down-counter underflows because refreshing of the down-counter is not possible due to the program having entered crashed execution or if a refresh error occurs due to refreshing outside the refresh-permitted period, the WDT outputs the reset signal or non-maskable interrupt request/interrupt request (WUNI).

After the reset signal or non-maskable interrupt request/interrupt request is output of for one cycle of counting, the value of the timeout period is set in the down-counter counting is restarted.

Reset output or interrupt request output can be selected by setting the OFS0.WDTRSTIRQS bit.

Figure 34.4 shows an example of operation (non-maskable interrupt) under the following conditions.

- Auto start mode (OFS0.WDTSTRT = 0)
- Non-maskable interrupt request output is enabled (OFS0.WDTRSTIRQS = 0)
- The window start position is 75% (OFS0.WDTRPSS[1:0] = 10b)
- The window end position is 25% (OFS0.WDTRPES[1:0] = 10b)

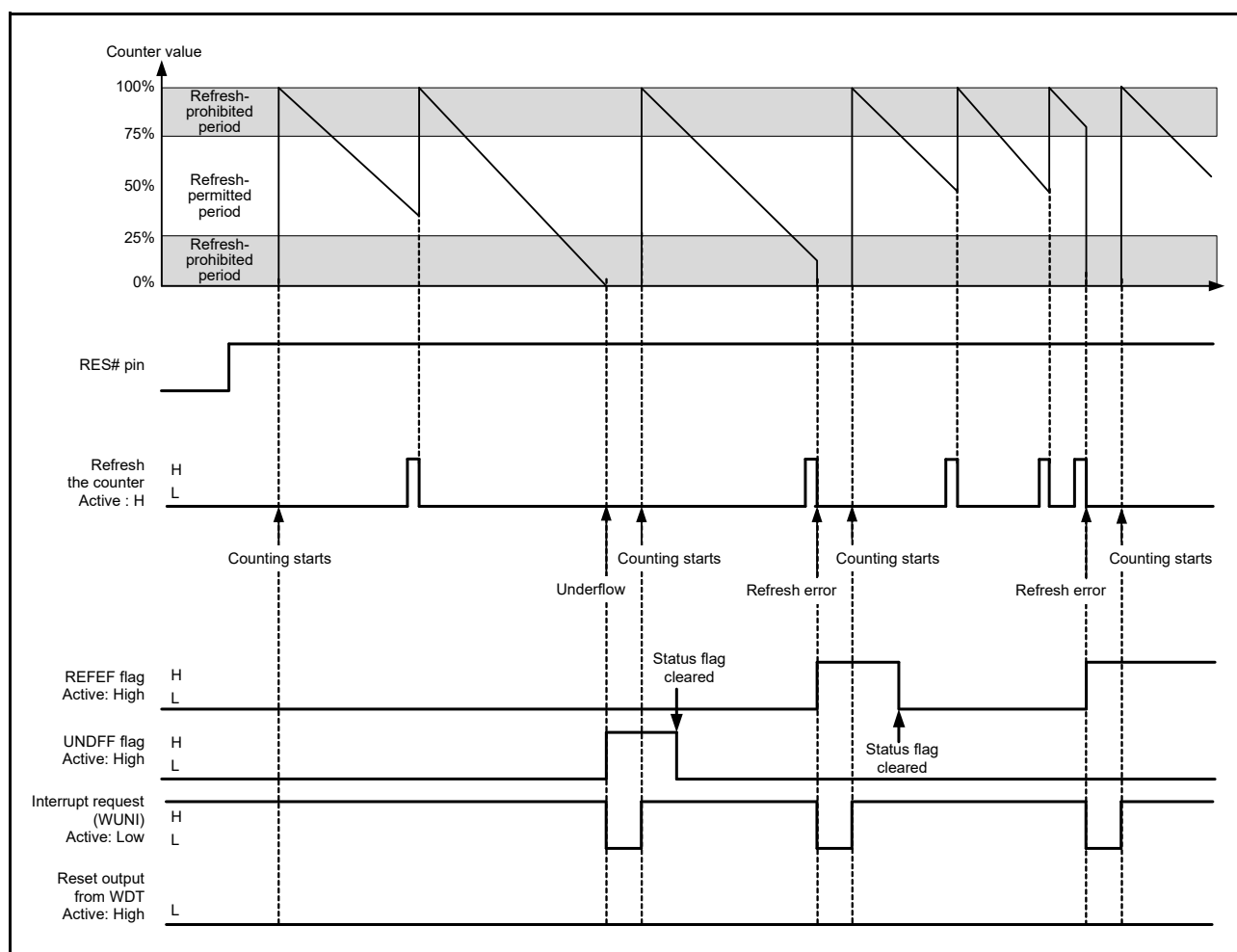


Figure 34.4 Operation Example in Auto-Start Mode

34.3.2 Control over Writing to the WDTCR and WDTRCR Registers

Writing to the WDTCR or WDTRCR register is only possible once between the release from the reset state and the first refresh operation.

After a refresh operation (counting starts) or by writing to the WDTCR or WDTRCR register, the protection signal in the WDT becomes 1 to protect the WDTCR and WDTRCR registers against subsequent attempts at writing.

This protection is released by the reset source of the WDT. With other reset sources, the protection is not released.

Figure 34.5 shows control waveforms produced in response to writing to the WDTCR register.

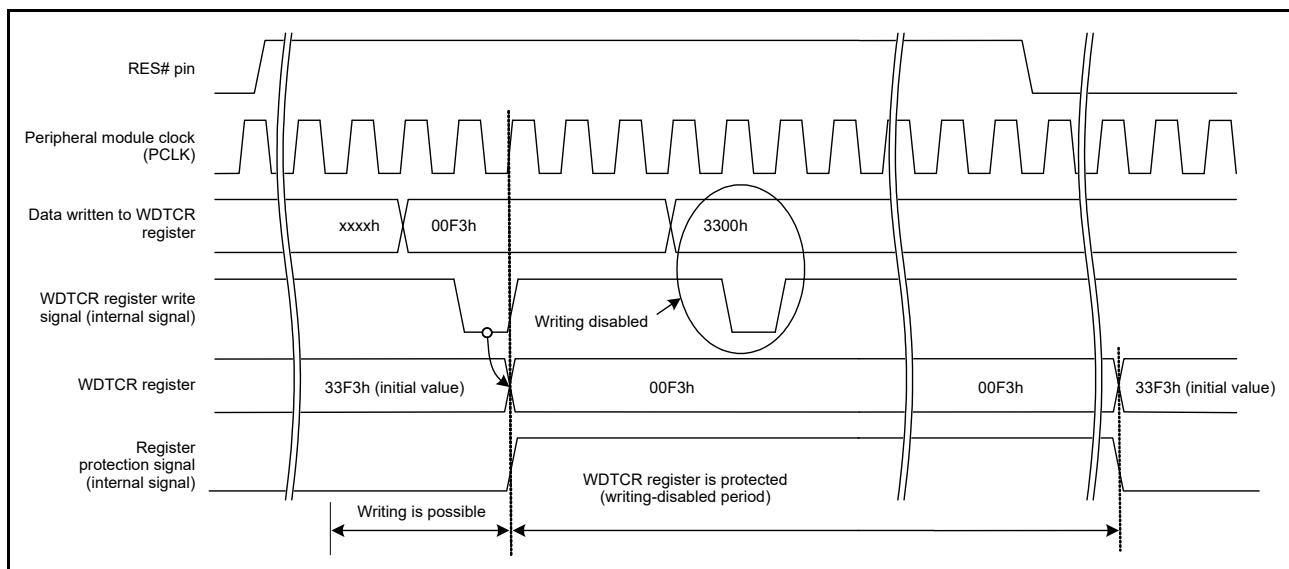


Figure 34.5 Control Waveforms Produced in Response to Writing to the WDTCR Register

34.3.3 Refresh Operation

The down-counter is refreshed by writing the values 00h and then FFh to the WDTRR register. If a value other than FFh is written after 00h, the down-counter is not refreshed. After such invalid writing, correct refreshing is performed by again writing to 00h and then FFh to the WDTRR register.

Even if a register other than the WDTRR register is accessed or the WDTRR register is read between writing 00h and writing FFh to the WDTRR register, correct refreshing will be done.

Writing to refresh the counter must be performed within the refresh-permitted period. Whether writing is done within the refresh-permitted period is determined when writing FFh. For this reason, correct refreshing will be done even if 00h is written outside the refresh-permitted period.

[Sample sequences of writing that are valid for refreshing the counter]

- 00h → FFh
- 00h (n-1-th time) → 00h (nth time) → FFh
- 00h → access to another register or read from the WDTRR register → FFh

[Sample sequences of writing that are not valid for refreshing the counter]

- 23h (a value other than 00h) → FFh
- 00h → 54h (a value other than FFh)
- 00h → AAh (00h and a value other than FFh) → FFh

After FFh is written to the WDTRR register, refreshing the down-counter requires up to four cycles of the signal for counting. Therefore, writing FFh to the WDTRR register should be completed four-count cycles before the down-counter underflows.

Figure 34.6 shows the WDT refresh-operation waveforms when the clock division ratio = PCLK/64.

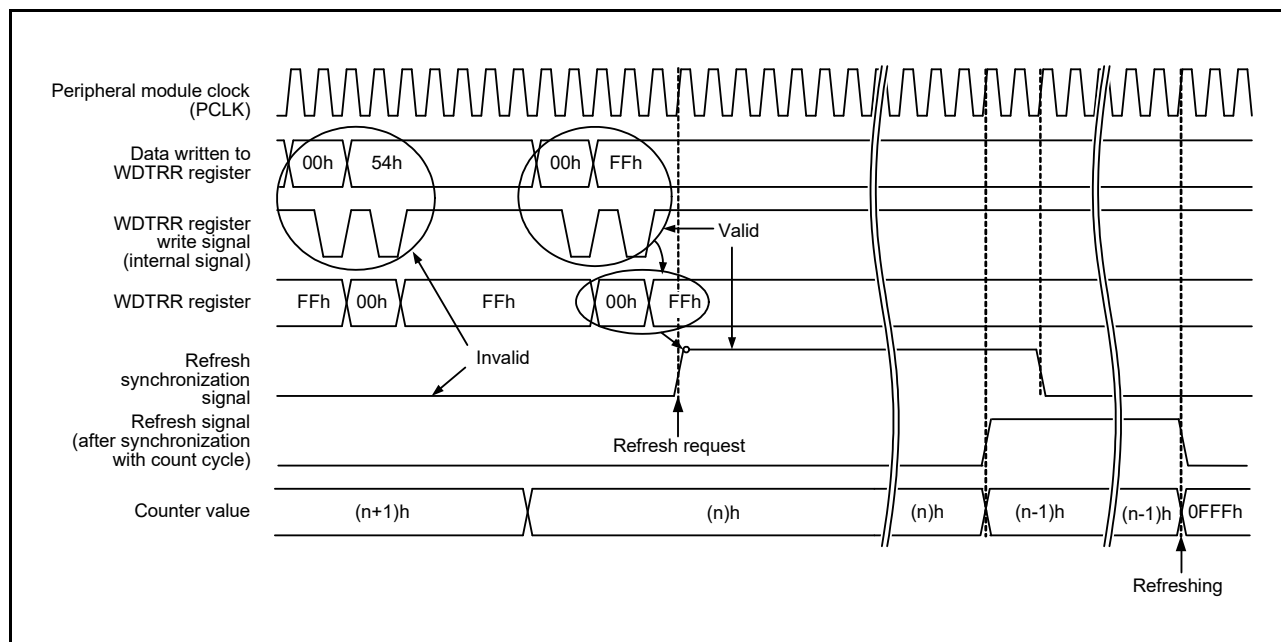


Figure 34.6 WDT Refresh Operation Waveforms (WDTCR.CKS[3:0] = 0100b, WDTCR.TOPS[1:0] = 01b)

34.3.4 Reset Output

When the WDTCR.RSTIRQS bit is set to 1 in register start mode or when the WDTRSTIRQS bit in option function select register 0 (OFS0) is set to 1 in auto-start mode, a reset signal is output for one-count cycle when an underflow in the down-counter or a refresh error occurs.

In register start mode, the down-counter is initialized (all bits set to 0) and stopped in that state after output of the reset signal. After the reset is released and the program is restarted, the counter is set up again and counting down is started by refreshing.

In auto-start mode, counting down automatically starts after the reset is released.

34.3.5 Interrupt Source

When the WDTCR.RSTIRQS bit is set to 0 in register start mode or when the OFS0.WDTRSTIRQS bit is set to 0 in auto-start mode, an interrupt (WUNI) signal is generated when an underflow in the counter or a refresh error occurs. This interrupt can be used as a non-maskable interrupt and an interrupt. For details, refer to section 15, Interrupt Controller (ICUD).

Table 34.4 WDT Interrupt Source

Name	Interrupt Source	DTC Activation	DMAC Activation
WUNI	Down-counter underflow Refresh error	Not possible	Not possible

34.3.6 Reading the Down-Counter Value

The WDT stores the counter value in the WDTSR.CNTVAL[13:0] bits. Thus, the counter value can be checked through the WDTSR.CNTVAL[13:0] bits.

Figure 34.7 shows the processing for reading the WDT down-counter value when the clock division ratio = PCLK/64.

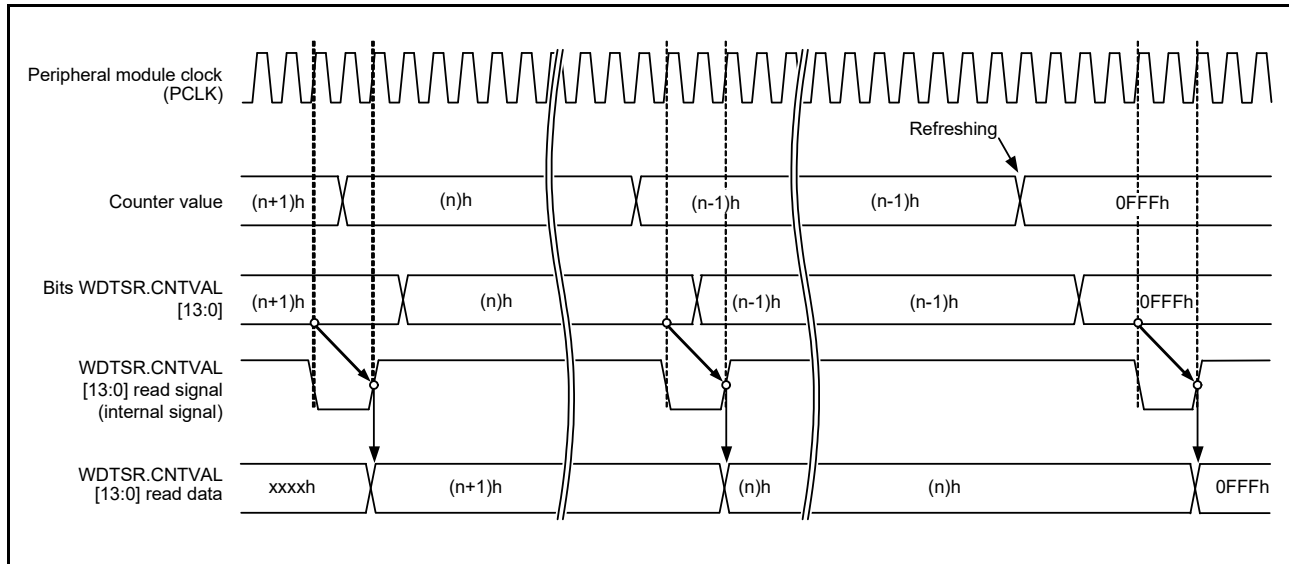


Figure 34.7 Processing for Reading WDT Down-Counter Value
(WDTCR.CKS[3:0] = 0100b, WDTCR.TOP[1:0] = 01b)

34.3.7 Correspondence between Option Function Select Register 0 (OFS0) and WDT Registers

Table 34.5 lists the correspondence between option function select register 0 (OFS0) used in auto-start mode and the registers used in register start mode.

Do not change the OFS0 register setting during WDT operation.

For details on the OFS0 register, refer to section 7.2.3, Option Function Select Register 0 (OFS0).

Table 34.5 Correspondence between Option Function Select Register 0 (OFS0) and WDT Registers

Target of Control	Function	OFS0 Register (Enabled in Auto-Start Mode) OFS0.WDTSTRT = 0	WDT Registers (Enabled in Register Start Mode) OFS0.WDTSTRT = 1
Down-counter	Timeout period selection	OFS0.WDTPSS[1:0]	WDTCR.TOP[1:0]
	Clock division ratio selection	OFS0.WDTCCKS[3:0]	WDTCR.CKS[3:0]
	Window start position selection	OFS0.WDTRPSS[1:0]	WDTCR.RPSS[1:0]
	Window end position selection	OFS0.WDTRPES[1:0]	WDTCR.RPES[1:0]
Reset output or interrupt request output	Reset output or interrupt request output selection	OFS0.WDTRSTIRQS	WDTCR.RSTIRQS